

Chapter 7: Work, Energy, and Energy Resources

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PHY201
Lecture 7



Part I

Work

Outline of Part I: Work

Definition of Work

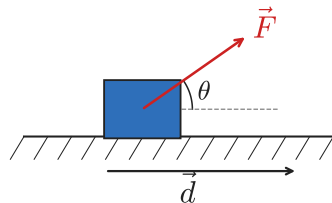
What Is Work?

Work is done on an object when a force acts on it *along* the direction of displacement.

$$W = \vec{F} \cdot \vec{d} = |\vec{F}| |\vec{d}| \cos \theta$$

- ▶ θ = angle between $\vec{F}$ and $\vec{d}$
- ▶ SI unit: **joule** ($J = N \cdot m$)
- ▶ W is a *scalar* (positive, negative, or zero)

$$W = Fd \cos \theta$$



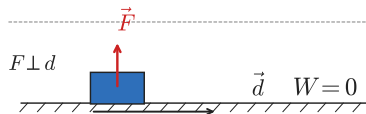
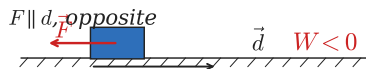
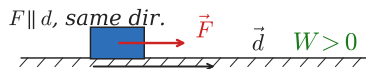
Positive, Negative, and Zero Work

Zero Work

If $\vec{F} \perp \vec{d}$, then $\cos 90^\circ = 0 \Rightarrow W = 0$.

Examples: normal force on a sliding block;
centripetal force on circular orbit; carrying a bag
horizontally.

Scenario	θ	Sign of W
$\vec{F}$ & $\vec{d}$ same direction	0°	+
$\vec{F}$ & $\vec{d}$ opposite	180°	−
$\vec{F}$ perpendicular to $\vec{d}$	90°	0



Work: Worked Examples

Bob pushes a lawnmower with $F = 100 \text{ N}$ at $\theta = 37.0^\circ$ below horizontal over $d = 2.50 \text{ m}$. How much work does Bob do?

$$\begin{aligned} W &= Fd \cos \theta \\ &= (100)(2.50) \cos(37.0^\circ) \end{aligned}$$

$$W = 199 \text{ J}$$

Check Your Understanding

In which of the following situations is **zero** net work done on the object by the listed force?

- (A) A horse pulls a cart forward along a flat road.
- (B) A waiter carries a tray horizontally across the room at constant velocity.
- (C) The Moon orbits Earth (gravity is the force).
- (D) A ball rolls along a frictionless floor (normal force).

Come to lecture to see the answer.

Check Your Understanding

A 50.0 kg rock climber ascends 5.00 m vertically.
How much work does **gravity** do on the climber?

Come to lecture to see the answer.

Part II

Kinetic Energy & the Work-Energy Theorem

Outline of Part II: Kinetic Energy & the Work-Energy Theorem

Kinetic Energy

Work-Energy Theorem

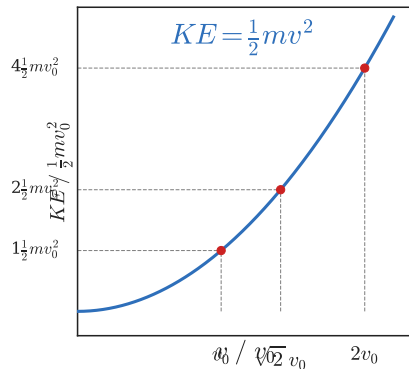
Kinetic Energy

Kinetic energy is the energy an object possesses due to its motion:

$$KE = \frac{1}{2}mv^2$$

- ▶ Always ≥ 0 (scalar, never negative)
- ▶ Doubles when $v \rightarrow \sqrt{2}v$; quadruples when $v \rightarrow 2v$
- ▶ Same units as work: J

KE depends on v^2 — a car at 60 mph has *four times* the KE of the same car at 30 mph.



Outline of Part II: Kinetic Energy & the Work-Energy Theorem

Kinetic Energy

Work-Energy Theorem

The Work-Energy Theorem

Work-Energy Theorem

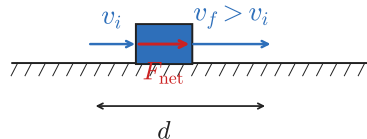
The *net* work done on an object equals the change in its kinetic energy:

$$W_{\text{net}} = \Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

 This is Newton's second law in terms of energy. $F_{\text{net}} = ma$ and kinematics ($v_f^2 = v_i^2 + 2ad$) together give $W_{\text{net}} = F_{\text{net}}d = \Delta KE$.

- ▶ $W_{\text{net}} > 0$: object speeds up
- ▶ $W_{\text{net}} < 0$: object slows down
- ▶ $W_{\text{net}} = 0$: constant speed

$$W_{\text{net}} = \Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$



Work-Energy Theorem: Baseball Example

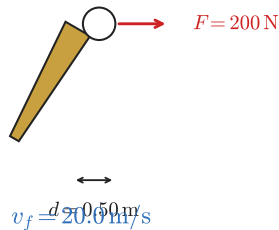
A 0.500 kg ball starts from rest. A bat exerts a net force of 200 N over a contact distance of 0.500 m. What is the ball's final speed?

$$W_{\text{net}} = F_{\text{net}} d = (200)(0.500) = 100 \text{ J}$$

$$W_{\text{net}} = \frac{1}{2} m v_f^2 - 0$$

$$v_f = \sqrt{\frac{2 W_{\text{net}}}{m}} = \sqrt{\frac{2(100)}{0.500}} = \boxed{20.0 \text{ m/s}}$$

$v_i = 0$ (at rest)



Check Your Understanding

A 1500 kg car traveling at 25.0 m/s brakes to a stop.
What is the net work done on the car by the braking force?

Come to lecture to see the answer.

Part III

Potential Energy & Conservation of Energy

Outline of Part III: Potential Energy & Conservation of Energy

Gravitational Potential Energy

Conservative vs. Non-Conservative Forces

Elastic Potential Energy

Conservation of Mechanical Energy

Gravitational Potential Energy

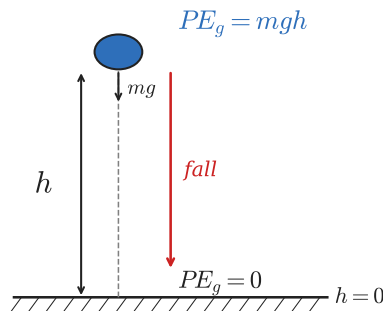
Gravitational PE is energy stored by position in a gravitational field (relative to a chosen reference height):

$$PE_g = mgh$$

- ▶ h measured from a reference point ($h = 0$)
- ▶ Only *changes* in PE_g matter physically
- ▶ Depends on mass, g , and height — not on path

Mass Cancels

The speed gained falling a height h is the *same* for all masses: $v = \sqrt{2gh}$, since $PE_g = mgh$ and $KE = \frac{1}{2}mv^2$ both scale with m .



Outline of Part III: Potential Energy & Conservation of Energy

Gravitational Potential Energy

Conservative vs. Non-Conservative Forces

Elastic Potential Energy

Conservation of Mechanical Energy

Conservative & Non-Conservative Forces

Conservative Forces

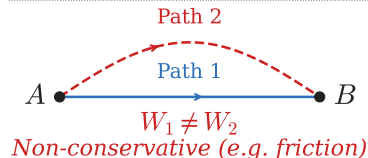
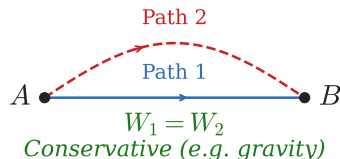
Work done is *path-independent*; energy can be stored and recovered.

- ▶ Gravity, spring (elastic) force
- ▶ Associated with a potential energy PE

Non-Conservative Forces

Work done *depends on path*; energy is permanently converted (often to heat).

- ▶ Friction, air drag, tension (when rope stretches)
- ▶ *Cannot* be associated with a PE



Outline of Part III: Potential Energy & Conservation of Energy

Gravitational Potential Energy

Conservative vs. Non-Conservative Forces

Elastic Potential Energy

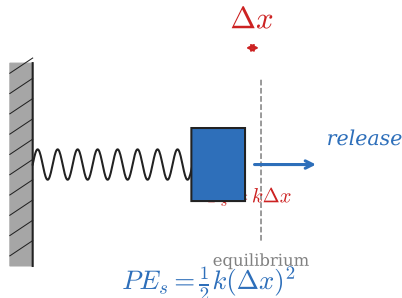
Conservation of Mechanical Energy

Elastic (Spring) Potential Energy

Energy stored in a compressed or stretched spring:

$$PE_s = \frac{1}{2}k(\Delta x)^2$$

- ▶ Δx = deformation from natural length
- ▶ Always ≥ 0 regardless of compression or extension
- ▶ Released as KE when spring returns to equilibrium



🚩 Recall Hooke's Law (Ch. 5)

$$\vec{F}_s = k\Delta\vec{x} \quad (\text{restoring force of a spring})$$

Outline of Part III: Potential Energy & Conservation of Energy

Gravitational Potential Energy

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Conservation of Mechanical Energy

Conservation of Mechanical Energy

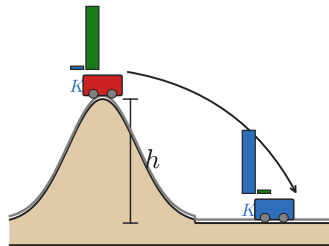
$$E_{\text{total}} = KE + PE$$

When only conservative forces act:

$$\Delta E_{\text{total}} = 0 \implies KE_i + PE_i = KE_f + PE_f$$

 Pick the state where one term is simplest (e.g. $v = 0$ or $h = 0$), set it as your reference, then equate total energies at two states.

$$KE_i + PE_i = KE_f + PE_f$$



Conservation of Energy: Ball Drop

A ball is dropped from rest at $h = 74.5$ cm.
What is its speed just before it hits the ground?

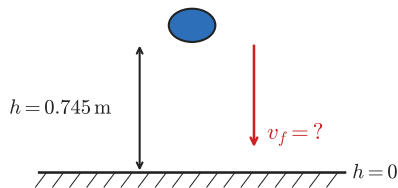
$$KE_i + PE_i = KE_f + PE_f$$

$$0 + mgh = \frac{1}{2}mv_f^2 + 0$$

$$v_f = \sqrt{2gh} = \sqrt{2(9.80)(0.745)}$$

$$v_f = 3.82 \text{ m/s}$$

$$mgh = \frac{1}{2}mv_f^2$$
$$v_f = \sqrt{2gh} = 3.82 \text{ m/s}$$



Note: mass cancels — result is independent of m .

Check Your Understanding

A roller-coaster car starts from rest at the top of a frictionless hill of height $h = 5.00$ m.

What is its speed at the bottom?

Come to lecture to see the answer.

Check Your Understanding

A spring with $k = 300 \text{ N/m}$ is compressed $\Delta x = 0.200 \text{ m}$ and launches a 0.120 kg block on a frictionless surface.

What is the block's speed after release?

Come to lecture to see the answer.

Part IV

Non-Conservative Forces & Energy Loss

Outline of Part IV: Non-Conservative Forces & Energy Loss

Work Done by Non-Conservative Forces

When NC Forces Are Present

General Energy Equation

Non-conservative forces change the total mechanical energy:

$$W_{\text{nc}} = \Delta E_{\text{total}} = \Delta KE + \Delta PE$$

Money Analogy

Think of E_{total} as your bank balance. Conservative forces move money between checking (KE) and savings (PE). Non-conservative forces (friction) are *taxes* — the money leaves the system permanently as heat.

- ▶ Friction: $W_{\text{nc}} = -f_k d < 0$
- ▶ Applied force: $W_{\text{nc}} = F_{\text{app}} d > 0$

NC Forces: Ramp Example

A 3.00 kg block slides down a 4.00 m ramp inclined at $\theta = 30.0^\circ$ from rest. Kinetic friction coefficient $\mu_k = 0.100$. Find the speed at the bottom.

$$h = (4.00) \sin 30.0^\circ = 2.00 \text{ m}$$

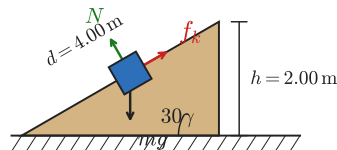
$$\begin{aligned} f_k &= \mu_k mg \cos \theta = 0.100(3.00)(9.80) \cos 30.0^\circ \\ &= 2.55 \text{ N} \end{aligned}$$

$$W_{\text{nc}} = -f_k d = -(2.55)(4.00) = -10.2 \text{ J}$$

$$W_{\text{nc}} = \Delta KE + \Delta PE$$

$$-10.2 = \frac{1}{2}(3.00)v_f^2 - (3.00)(9.80)(2.00)$$

$$v_f = 5.69 \text{ m/s}$$



Check Your Understanding

A 2.00 kg block starts from rest on a flat surface with $\mu_k = 0.300$. A horizontal force $F = 30.0$ N pushes it 5.00 m. What is its final speed?

Come to lecture to see the answer.

Part V

Power & Efficiency

Outline of Part V: Power & Efficiency

Power

Efficiency

Power

Power is the rate at which work is done (or energy is transferred):

$$P = \frac{W}{t} = \frac{E}{t}$$

- ▶ SI unit: **watt** ($W = J/s$)
- ▶ $1 \text{ hp} = 746 \text{ W}$ (historical unit)
- ▶ Also: $P = Fv$ (when force is along velocity)

Device	Typical Power
LED light bulb	10 W
Hair dryer	1500 W
Car engine	150 kW
Large power plant	1 GW

Check Your Understanding

A 600 W electric motor lifts a 50.0 kg crate straight up for 30.0 s (assume 100% efficiency).

How high does it lift the crate?

Come to lecture to see the answer.

Outline of Part V: Power & Efficiency

Power

Efficiency

Efficiency

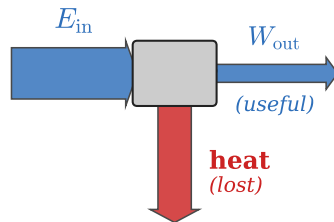
Definition

Efficiency measures how much of the input energy becomes useful output:

$$\eta = \frac{W_{\text{out}}}{E_{\text{in}}} \times 100\%$$

Always $\eta \leq 100\%$ (NC forces like friction always steal some energy).

$$\eta = \frac{W_{\text{out}}}{E_{\text{in}}} \times 100\%$$



Device/Activity	η
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Human cycling	$\sim 25\%$
Car engine (gasoline)	$\sim 25\%$
LED bulb	$\sim 90\%$
Electric motor	$\sim 95\%$

Check Your Understanding

A machine receives $E_{\text{in}} = 5000 \text{ J}$ of energy and does $W_{\text{out}} = 1500 \text{ J}$ of useful work.

What is its efficiency?

Come to lecture to see the answer.

Next Lecture

Chapter 8: Linear Momentum and Collisions

- ▶ Momentum $\vec{p} = m\vec{v}$ and impulse
- ▶ Conservation of momentum
- ▶ Elastic and inelastic collisions
- ▶ Center of mass

Key connection: Just as the work-energy theorem links force to energy, the impulse-momentum theorem links force to momentum. Both are consequences of $F = ma$.